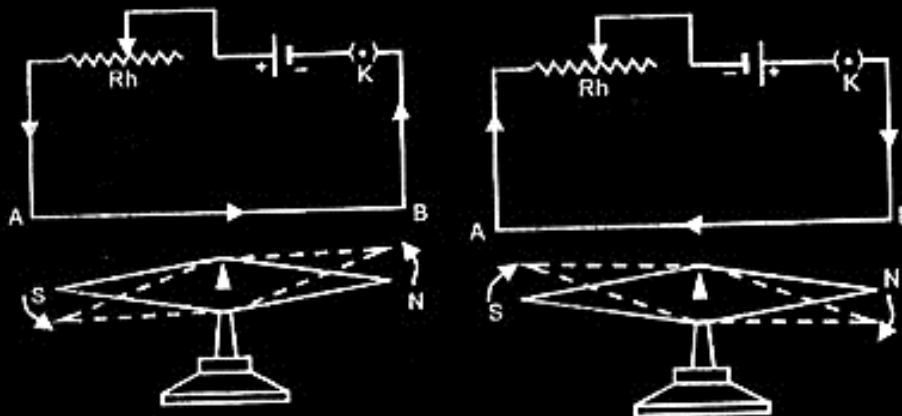


Oersted's Experiments: In 1819 a Danish scientist named Hans Christian Oersted discovered a close relationship between electricity and magnetism with this experiment. The new knowledge led to the study of electromagnetism.

According to the experiments, a conducting wire is kept along the length of a freely placed magnetic needle and a bit above it. If electric current is passed through the wire, it is found that magnetic needle is deflected from its original position. The magnetic needle tries to align itself at right angles to the conductor. But the needle does not align itself at right angles, because there is also the influence of earth's magnetism. As a result, due to the combined influence of magnetic field produced by the electric current and earth's magnetic field, the needle is deflected from the magnetic meridian and remain in equilibrium position.



If current is increased in the conductor, then the deflection of the needle also increases. If the direction of the current is reversed in the conductor, in that case the magnetic needle also deflects but in opposite direction to the earlier.

Again, if this experiment is performed keeping the wire below the needle, the needle also deflects. Keeping the direction of the current same, if the conductor is placed below the needle, then the direction of the rotation of the needle would be opposite if the conductor is placed above the needle. If the current is stopped in the conductor, then the needle returns to its original position.

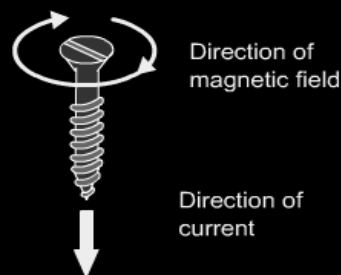
The following conclusions can be drawn from the Oersted's Experiment:

1. When current flows through a conductor magnetic field is produced around it.
2. Magnitude of the magnetic field depends on the strength of current.
3. The direction of magnetic field depends on the direction of the current flow.
4. Magnetic field exists as long as the flow of current exists in the wire.

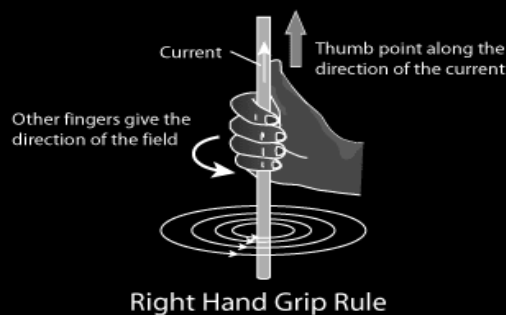
Direction of the magnetic field produced by electric current:

The direction of the magnetic field can be found using any of the following law:

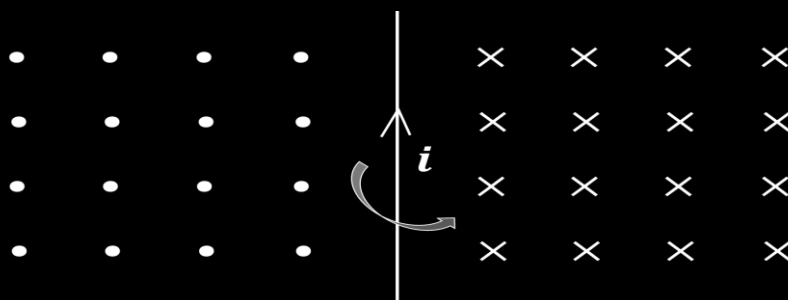
1. **Maxwell's cork screw law:** If a right handed screw is rotated along a current carrying conductor in the direction of the current, then the direction of the magnetic field will be in the direction in which the thumb rotates, i.e., the north pole of the magnetic needle will be deflected in this direction.



2. **Fleming's right hand rule:** If a current carrying conductor is grasped with the palm of the right hand so that the thumb point the direction of the current, then the tips of the other fingers will indicate the direction of the magnetic field.



Representation of Magnetic field in plane of paper: According to the Fleming's right hand rule, the dot sign ($\bullet$) is used to represent the magnetic field outward the plane or towards the reader and the cross sign ($\times$) represents inward direction of the field, as shown in figure below:



Magnetic Force: We know that a stationary charged particle produced electric field around it. Similarly we observed that a moving charge produced a magnetic field around it. Beside the electric Coulombs force, a moving charge exerts other force on another moving charge. This velocity dependent force on charges is known as magnetic force.

From experiment, it is found that the magnitude of the force (F) acting on a magnet charge in a magnetic field is –

- (a) directly proportional to the magnitude of the charge (q).
- (b) directly proportional to the velocity (v) of the charge.
- (c) directly proportional to the magnitude of the magnetic field (B)
- (d) directly proportional to the *sine* of the angle (θ) between the direction of the velocity of the charge and the direction of the magnetic field.

Thus, $F \propto qvB\sin\theta$

In S.I. unit the constant of proportionality is 1. So,

$$F = qvB\sin\theta$$

$$\vec{F} = q (\vec{v} \times \vec{B})$$

Magnetic force on the current carrying wire is –

$$F = ilB\sin\theta$$

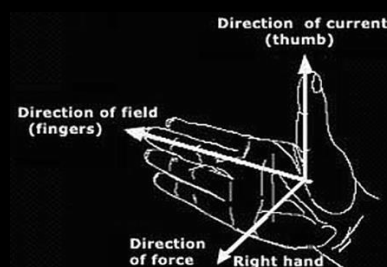
Where, i = Current flow through the conducting wire

l = Length of the wire

B = Magnetic field

θ = Angle between current flow direction and magnetic field.

Fleming's right hand rule – 1: When the right hand is stretched, if the fingers indicate the magnetic field and the thumb indicates the velocity of the charge or current flow direction, then outward normal drawn on the palm will indicate the direction of force for the positive charge or on the current carrying wire.



Some definitions:

- **Magnetic field:** The space or region around a magnet or a current carrying wire whereby the magnetic needle shows the deflection, that space or region is called magnetic field of that magnet or current carrying wire.
- **Magnetic flux:** Magnetic field is generally represented by magnetic field lines. If magnetic field is strong then lines are closer and density populated.

"Number of field lines passing through either a real or a fictitious plane in a magnetic field is called magnetic flux (ϕ)"

S.I. unit of magnetic flux is *Weber (Wb)* or NmA^{-1} .

1weber: Number of flux that changes per second in a coil for the production of 1volt potential difference in that coil is called 1Weber.

- **Magnitude of the Magnetic field (B):**

If a unit positive charge moving perpendicular to the magnetic field with unit velocity, then the force experienced by the charge is called the magnitude of the magnetic field.

$$B = \frac{F}{vq}$$

On the other hand, the number of magnetic field lines or magnetic flux incident normally on unit area of a real or fictitious plane is called flux density in that plane. It is called magnetic field or magnetic induction. i.e.,

$$B = \frac{\phi}{A}$$

S.I. unit of it is Wbm^{-2} or *Tesla(T)* or $NA^{-1}m^{-1}$. Also, $1Tesla = 10^4 Gauss$.

1Tesla: If 1Coulomb of charge is in motion perpendicular to direction of magnetica field with a velocity of $1ms^{-1}$ and experience a force of 1N, then the magnitude of that magnetic field is called 1Tesla (T).

Lorentz Force: The force that a charge experiences while moving simultaneously through the electric field and magnetic field is called Lorentz force.

If $\vec{B}$ is the magnetic field and $\vec{E}$ is the electric field, then the Lorentz force for the electron passing through both these fields is –

$$F = F_e + F_m$$

$$F = qE + qvB\sin\theta$$

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

Problem-1: An electron is moving with a speed of $5 \times 10^7 \text{ m/s}$ at right angle to a magnetic field of 5000 G . (a) What is the magnetic force on the electron? (b) What is the radius of circle in which the electron moves?

Problem-2: When an electron moves with a velocity 10^5 ms^{-1} making an angle of 60° with a magnetic field of 0.50 T then find the magnitude of the force acting on the electron.

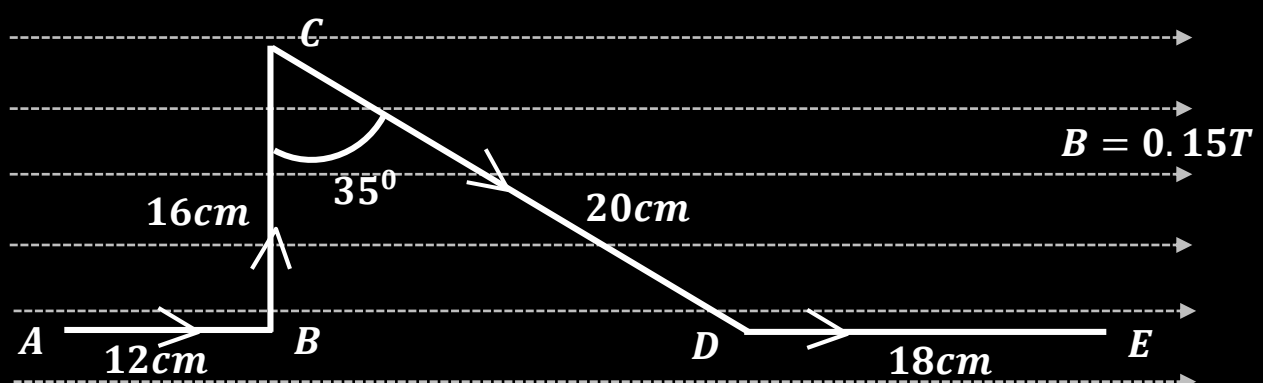
Problem-3: When a proton moves with a velocity $2 \times 10^6 \text{ ms}^{-1}$ making an angle of 30° with a magnetic field at a certain place it experiences of force of $4.8 \times 10^{-15} \text{ N}$. Find the magnitude of the magnetic field.

Problem-4: When a 10 cm long wire carrying 10 A current is at an angle of 30° with a magnetic field then it experiences 5 N force. Find the magnitude of the magnetic field.

Problem-5: At the equator, the earth's magnetic field is nearly horizontal, directed from the southern to northern hemisphere. Its magnitude is about 0.50 G . Find the force on a 20 m wire carrying a current of 30 A parallel to the earth (a) from east to west (b) from north to south.

Problem-6: A straight horizontal segment of copper wire carries $i = 28 \text{ A}$. What are the magnitude and direction of the magnetic field needed of float the wire, that is, to balance its weight? Its linear mass density is 46.6 g/m .

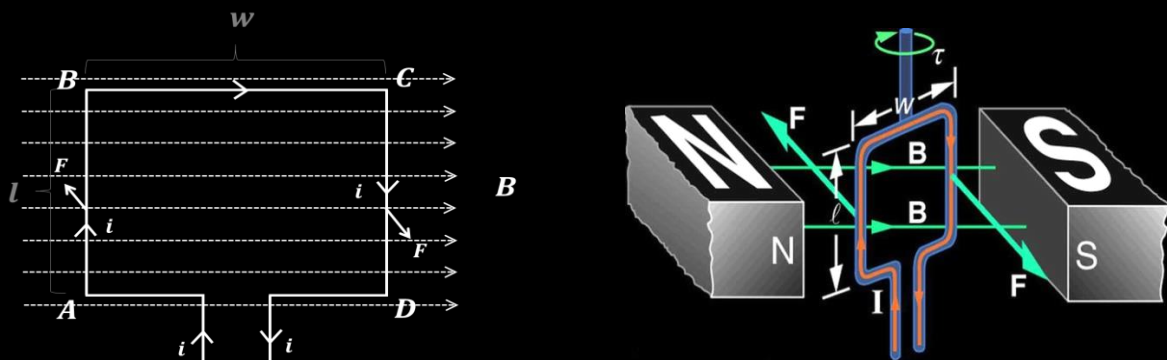
Problem-7: Find the force on each segment of the wire shown in figure below, if $B = 0.15 \text{ T}$. Assume that the current in the wire is 5 A .



Torque on a Current Carrying Coil in a Magnetic field:

We know that if a current carrying wire is placed in a magnetic field, a magnetic force acts on it. From this phenomenon, when a current carrying loop or coil is placed in a magnetic field, magnetic force and torque produced due to the current flow in the coil.

Let $ABCD$ be very small rectangular loop or coil carrying current placed in a uniform magnetic field B . The plane of the coil is in parallel position to the magnetic field. The loop is attached to a shaft or rod in such a way as it can rotate freely as shown in figure.



As the loop is rectangular –

$$AB = CD = l \quad (\text{length of the loop})$$

and $BC = DA = w \quad (\text{width of the loop})$

Let the current flow, i is clockwise. Now the force applied on the arms –

$$AB \rightarrow F = ilB\sin 90 = ilB$$

$$BC \rightarrow F = iwB\sin 0 = 0$$

$$CD \rightarrow F = ilB\sin 90 = ilB$$

$$DA \rightarrow F = iwB\sin 180 = 0$$

According to the right hand rule, the forces acting on AB and CD are equal and opposite to each other and their line of action not being in same straight line, the loop will experience a torque or torsional force; consequently the loop will turn or rotate with respect to the rod. Now – Net torque = Force $\times$ perpendicular distance between the forces acting

$$\tau = F \times w$$

$$\tau = ilB \times w = ilwB$$

$$\tau = iAB$$

Here, $A = lw$, the area of the loop.

If the coil is made by warpping N number of turns of a wire, then –

$$\tau = NiAB$$

If the plane of the loop or coil makes an angle θ with the magnetic field, then –

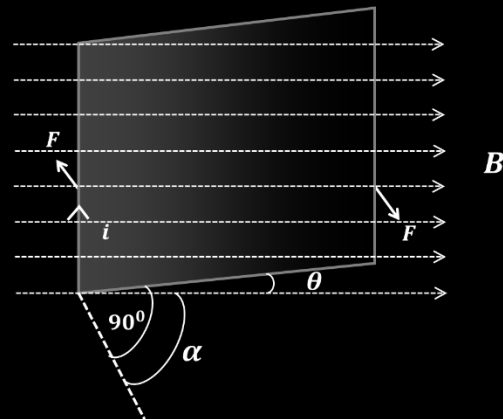
$$\tau = NiAB\cos\theta$$

Again, If the normal of the plane (Area vector, $\vec{A}$) of the loop or coil makes an angle α with the magnetic field, then –

$$\tau = NiAB\sin\alpha$$

In the vector form it can be written as –

$$\tau = Ni(\vec{A} \times \vec{B})$$



$Ni\vec{A}$ is called the magnetic moment, $\vec{M}$ of the coil. The direction of $\vec{M}$ will be along the direction of $\vec{A}$. Now –

$$\tau = \vec{M} \times \vec{B}$$

Magnetic Moment, $\vec{M}$: The product of the current of a current carrying coil and area of the coil is called the magnetic moment of that coil.

Problem-1: An electron is moving with a speed of $5 \times 10^7 \text{ m/s}$ at right angle to a magnetic field of 5000G. What is the magnetic force on the electron?

Problem-2: A coil of 20 turns has an area of 800 mm^2 and bears a current of 0.5A. It is placed with its plane parallel to magnetic field of intensity 0.3T. Determine the torque and magnetic moment on the coil.

Problem-3: Current flowing in a circular coil of radius of $5 \times 10^{-2} \text{ m}$ having number of turns 100 is 1 ampere. If it is placed in a magnetic field of $1.5 \times 10^{-2} \text{ wb m}^{-2}$ at an angle of 30° between normal of the plane of the coil and the magnetic field, what magnitude of torque will be applied on the coil?

Problem-4: See the book “Physics for Engineers (Part-2) - Gias Uddin Ahmed”; Page-214

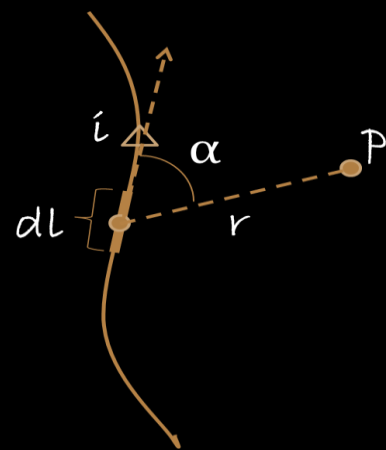
The Biot-Savart Law:

An electric current flowing in a conductor, or a moving electric charge, produces a magnetic field, or a region in the space around the conductor in which magnetic forces may be detected. The value of the magnetic field at a point in the surrounding space may be considered the sum of all the contributions from each small element, or segment, of a current-carrying conductor. The Biot-Savart law states how the value of the magnetic field at a specific point in space from one short segment of current-carrying conductor depends on each factor that influences the field.

According to the Biot-Savart Law –

The magnitude of magnetic field (dB) at a point produced around a conductor of small length due to the flow of current through it is –

- i) directly proportional to the amount of current. i.e., $dB \propto i$
- ii) directly proportional to the length of the conductor. i.e., $dB \propto dl$
- iii) directly proportional to the *sine* of the angle between the lines connecting the point from the mid-point of the conductor and the conductor itself. i.e., $dB \propto \sin\alpha$
- iv) inversely proportional to the distance between the point and the mid-point of the conductor. i.e., $dB \propto \frac{1}{r^2}$



So that,
$$dB \propto \frac{idl \sin\alpha}{r^2}$$

$$\Leftrightarrow dB \propto \frac{idl \sin\alpha}{r^2}$$

$$\Leftrightarrow dB = K \frac{idl \sin\alpha}{r^2}$$

Here, K is a proportionality constant whose value depends on the nature of the medium and on the units of the quantities.

In S.I. unit for air medium or vacuum –

$$K = \frac{\mu_0}{4\pi}; \text{ Here, } \mu_0 \text{ is the magnetic permeability of vacuum. } \mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

Now, putting the value of K , then –

$$dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin \alpha}{r^2}$$

For other medium –

$$dB = \frac{\mu}{4\pi} \times \frac{idl \sin \alpha}{r^2}$$

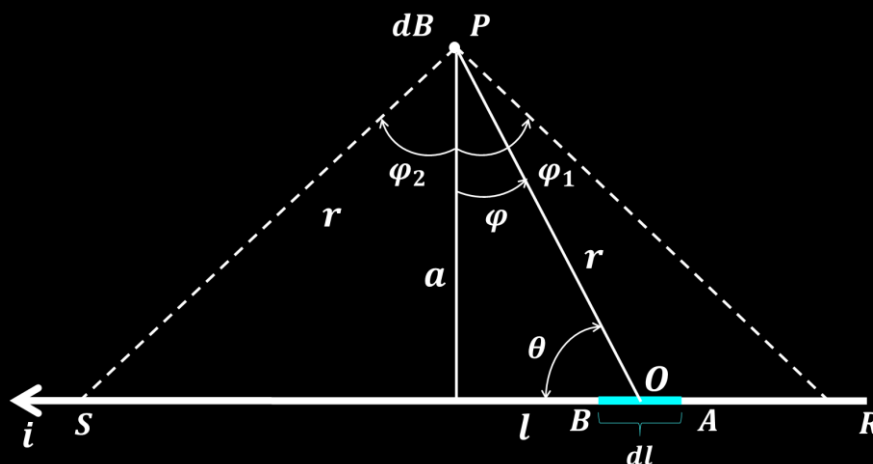
Vector form of the law –

$$\vec{dB} = \frac{\mu i}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

Applications of Biot-Savart Law:

(a) Magnetic field due to a straight current carrying wire of infinite length:

Suppose RS is a long and straight wire and i ampere current is flowing through it. Magnetic induction or magnetic field at point P at a distance r from the wire is to be determined [Figure]



Let a very small section AB of that wire be taken. Let its length be dl . Suppose the distance from the center point O of that portion to be point P is r and angle $\angle POB = \theta$.

According to the Biot-Savart's law, the magnetic field at point P –

$$dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin \theta}{r^2}$$

Now from figure, we get –

$$\sin \theta = \frac{a}{r} = \cos \phi$$

and

$$r = \frac{a}{\cos\varphi}$$

Also –

$$\tan\varphi = \frac{l}{a}$$

$$\Leftrightarrow l = a \tan\varphi$$

$$\therefore dl = a \sec^2\varphi d\varphi$$

Substituting r , dl and $\sin\theta$ in dB –

$$dB = \frac{\mu_0 i}{4\pi} \frac{a \cos\varphi \sec^2\varphi d\varphi}{\frac{a^2}{\cos^2\varphi}}$$

$$\therefore dB = \frac{\mu_0 i}{4\pi} \frac{\cos\varphi d\varphi}{a}$$

Magnetic field due to a whole conductor is obtained by integrating with limit from $-\varphi_1$ to φ_2 .
 (φ_1 is taken negative since it is Anticlockwise)

$$B = \frac{\mu_0 i}{4\pi a} \int_{-\varphi_1}^{\varphi_2} \cos\varphi d\varphi$$

$$\Leftrightarrow B = \frac{\mu_0 i}{4\pi a} [\sin\varphi]_{-\varphi_1}^{\varphi_2}$$

$$\Leftrightarrow B = \frac{\mu_0 i}{4\pi a} [\sin\varphi_2 - \sin(-\varphi_1)] = \frac{\mu_0 i}{4\pi a} [\sin\varphi_2 + \sin\varphi_1]$$

The wire is taken as infinite long, so that, $\varphi_1 = \varphi_2 \cong 90^\circ$; then –

$$B = \frac{\mu_0 i}{4\pi a} [\sin 90^\circ + \sin 90^\circ] = \frac{\mu_0 i}{4\pi a} [1 + 1]$$

$$\therefore B = \frac{\mu_0 i}{2\pi a}$$

Magnetic field intensity produced due to a current carrying wire does not depend on the surrounding medium.

(b) Magnetic field due to a circular current loop:

Figure shows a circular loop of radius R carrying a current i , we would like to calculate the magnetic field B at a point P on the axis of the loop a distance Z from the center of the loop.

Let us first consider a small element dl as shown in the figure. The magnitude of the magnetic field dB produced by this element at the point P is given by Biot-Savart's law to be –

$$dB = dB = \frac{\mu_0}{4\pi} \times \frac{idl \sin 90^\circ}{r^2}$$

Where, the angle between the element dl and r is 90° .

dB may be resolved into two components, one, $dB_{||}$ along the axis of the loop, and another is the perpendicular component $dB_{\perp}$. The perpendicular component $dB_{\perp}$ for every element dl will be equal and opposite to each other and the sum of all $dB_{\perp}$ for the complete loop is zero. Only the components along the axis add and the total magnetic field B at point P is simply the sum of all $dB_{||}$. i.e.,

$$B = \int dB = \int dB_{||}$$

$$\text{Now, } dB_{||} = dB \cos \alpha = \frac{\mu_0 i}{4\pi} \frac{dl \cos \alpha}{r^2}$$

Hence,

$$B = \int dB_{||} = \int \frac{\mu_0 i}{4\pi} \frac{dl \cos \alpha}{r^2}$$

From figure, the relationships are –

$$r = \sqrt{R^2 + Z^2}$$

$$\text{and } \cos \alpha = \frac{R}{r} = \frac{R}{\sqrt{R^2 + Z^2}}$$

Substituting these values and we get,

$$B = \int \frac{\mu_0 i dl}{4\pi(R^2 + Z^2)} \frac{R}{\sqrt{R^2 + Z^2}} = \frac{\mu_0 i R}{4\pi(R^2 + Z^2)^{3/2}} \int_0^{2\pi R} dl$$

$$\Rightarrow B = \frac{\mu_0 i R}{4\pi(R^2 + Z^2)^{3/2}} 2\pi R$$

$$B = \frac{\mu_0 i R^2}{2(R^2 + Z^2)^{3/2}}$$

An important special case of this result is the magnetic field at the center of the loop ($z = 0$). Equation then reduces to –

$$B = \frac{\mu_0 i}{2R}$$

For N turns of the circular wire –

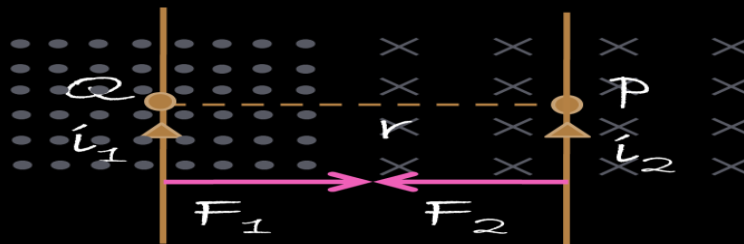
$$B = \frac{\mu_0 N i}{2R}$$

At distances from the loop, Z is much greater than R . R^2 can, therefore, be neglected compared to Z^2 in the denominator of the main equation then becomes –

$$B = \frac{\mu_0 i R^2}{2Z^3}$$

Forces between two parallel current carrying conductors:

Figure shows two long parallel wires Q & P separated by distance r . Wire Q carrying current i_1 and produces a magnetic field and the second wire P carrying current i_2 and also produces a magnetic field. According to the Fleming's right hand rule, the wire P will be experienced the force toward the wire Q and vice-versa the Q wire will be experienced a force toward the wire P .



The magnitude of the forces – $F_2 = \frac{\mu_0 i_1 i_2 l_2}{2\pi r}$ and $F_1 = \frac{\mu_0 i_1 i_2 l_1}{2\pi r}$

The forces per unit length will be equal to each other.

$$\frac{F_1}{l_1} = \frac{F_2}{l_2} = F' = \frac{\mu_0 i_1 i_2}{2\pi r}$$

The important conclusion of these parallel current carrying wire is –

- For parallel current direction of both wire, attract each other
- For anti-parallel current direction of both wire, repel each other

Ampere's Law:

By this law of the magnitude i.e., the magnetic field intensity produced due to the current flow through a conductor can be determined. It establishes the relation between the current i flowing through that conductor and the magnetic field $\vec{B}$ produced in it. The law is as follows:

“The line integral of a magnetic field along a close path is equal to μ_0 times, the net current flowing through the area subtended by the path”

That means –

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

Here, μ_0 = magnetic permeability of vacuum, dl = displacement vector of path difference, $\oint$ = integration in a close path.

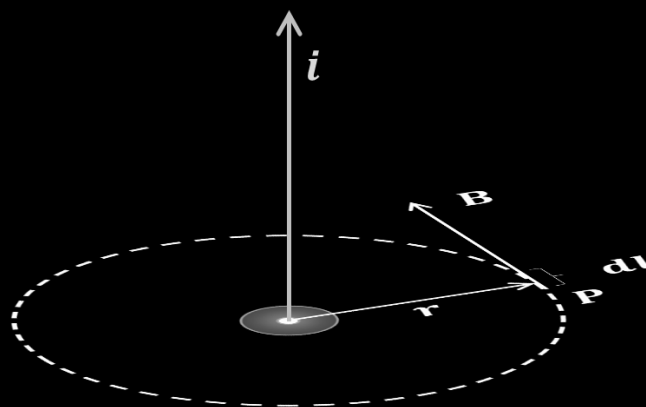
If the closed path does not link with the current loop, then –

$$\oint \vec{B} \cdot d\vec{l} = 0$$

Applications of Ampere's Law: It can be applied to determine the magnetic field B –

a) For the long straight current carrying wire:

Let us consider a long straight wire is carrying current i . We want to find the magnetic field at a point P a distance r from the wire. Figure shows the geometry for the calculation.



The lines of magnetic induction for a long straight wire carrying a current i are concentric circles centered on the wire. Hence the point P may be regarded to be on a circular loop of the radius r surrounding the wire. Applying Ampere's law, we have –

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\Leftrightarrow \oint B dl \cos \theta = \mu_0 i$$

Where, θ is the angle between the directions of B and dl . Now B has the same magnitude on any point on the circle. Moreover, B and dl point in the same direction, i.e., $\theta = 0^\circ$, then –

$$B \oint dl = \mu_0 i$$

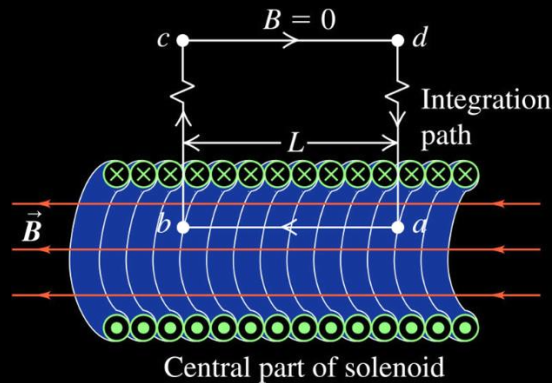
$$\Leftrightarrow B \cdot 2\pi r = \mu_0 i$$

$$\therefore B = \frac{\mu_0 i}{2\pi r}$$

This equation is similar to that obtained by Biot-Savart law for long straight wire.

b) For a long Solenoid:

Figure shows the section of the solenoid, the dots indicating that current in the wire is outward from and the crosses inward to the page.



Application of Ampere's law to the rectangular path $abcd$ in the direction shown in the figure gives –

$$\oint B \cdot dl = \oint_a^b B \cdot dl + \oint_b^c B \cdot dl + \oint_c^d B \cdot dl + \oint_d^a B \cdot dl$$

Now, since B is perpendicular to the sides bc and da the second and fourth integrals on the left side do not contribute. Also, B vanishes outside the solenoid. Hence the third integral also contributes nothing. The net contribution comes only from the first integral. For this path, B is parallel to dl and constant at all points. Thus –

$$\oint B \cdot dl = \oint_a^b B \cdot dl = B \int_a^b dl = BL$$

Where, L is the length of the path ab . The net current I that passes through the area bounded by the path $abcd$ is not the same as the current I_0 in the solenoid. Let n be the number of turns per unit length of the solenoid. Then, $I = I_0 nL$

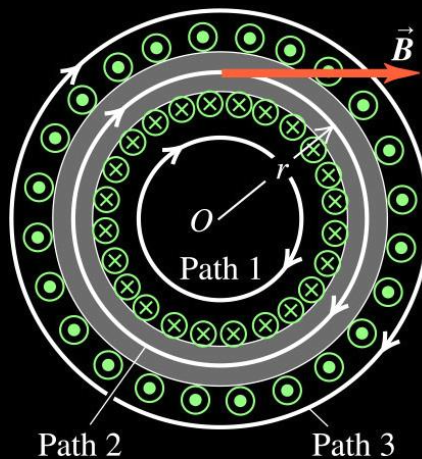
Ampere's law then becomes –

$$BL = \mu_0 I = \mu_0 I_0 nL$$

$$B = \mu_0 I_0 n$$

b) For a Toroid:

Figure shows the cross-section of a toroid. Again dots and crosses denoted the direction of the current in the wire of the toroid. From symmetry, the lines of B form concentric circles inside the toroid, as shown in the figure.



Application of Ampere's law to the circular path of the radius r , gives –

$$B \oint dl = \mu_0 I$$

$$\Leftrightarrow B \cdot 2\pi r = \mu_0 I$$

$$\therefore B = \frac{\mu_0 I}{2\pi r}$$

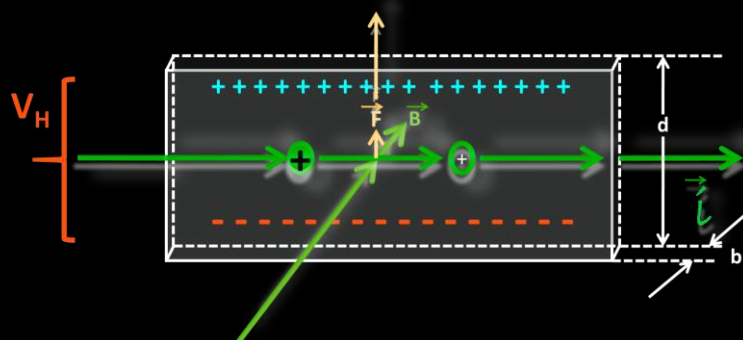
If I_0 is the current in the toroid wire and N is the total number of the windings, then –

$$B = \frac{\mu_0 N I_0}{2\pi r}$$

Hall Effect:

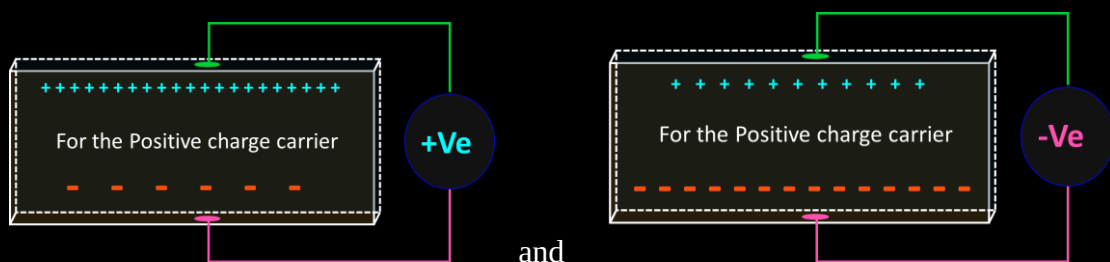
The Hall Effect is the production of a voltage difference (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and to an applied magnetic field perpendicular to the current. It was discovered by Edwin Hall in 1879.

If an electric current flows through a conductor in a magnetic field, the magnetic field exerts a transverse force on the moving charge carriers which tends to push them to one side of the conductor. This is most evident in a thin flat conductor as illustrated. A buildup of charge at the sides of the conductors will balance this magnetic influence, producing a measurable voltage between the two sides of the conductor. The presence of this measurable transverse voltage is called the Hall Voltage.



Applications of Hall Effects: From the Hall voltage, it can be determined that –

(a) **The nature of the charges:** (for detail follow the recommended books)



(b) **The number of charges per unit volume:** (for detail follow the recommended books)

The equation of the Hall voltage:

$$V_H = \frac{BI}{ntq}$$

(Some recommended mathematical problem should be solved)

Problem 1: A solenoid is 1.0m long and 3.0cm in mean diameter. It has five layers of windings of 850 turns each and carries current of 5.0amp . (a) What is B at its centre? (b) What is the magnetic flux ϕ_B for a cross-section of the solenoid at this centre?

Problem 2: An air-core solenoid with 2000 loops on it is 600mm long and has diameter of 20mm . If a current of 5A is sent through it, what will be the flux density within?

Problem 3: A solenoid 0.5m long has 2000 turns. The magnetic induction near the centre of the solenoid is 0.08T . What is the current in the solenoid?

11.7: The Hall Effect

Learning Objectives

By the end of this section, you will be able to:

- Explain a scenario where the magnetic and electric fields are crossed and their forces balance each other as a charged particle moves through a velocity selector
- Compare how charge carriers move in a conductive material and explain how this relates to the Hall effect

In 1879, E.H. Hall devised an experiment that can be used to identify the sign of the predominant charge carriers in a conducting material. From a historical perspective, this experiment was the first to demonstrate that the charge carriers in most metals are negative.

Visit this [website](#) to find more information about the Hall effect.

We investigate the **Hall effect** by studying the motion of the free electrons along a metallic strip of width l in a constant magnetic field (Figure 11.7.1). The electrons are moving from left to right, so the magnetic force they experience pushes them to the bottom edge of the strip. This leaves an excess of positive charge at the top edge of the strip, resulting in an electric field E directed from top to bottom. The charge concentration at both edges builds up until the electric force on the electrons in one direction is balanced by the magnetic force on them in the opposite direction. Equilibrium is reached when:

$$eE = ev_d B \quad (11.7.1)$$

where e is the magnitude of the electron charge, v_d is the drift speed of the electrons, and E is the magnitude of the electric field created by the separated charge. Solving this for the drift speed results in

$$v_d = \frac{E}{B}. \quad (11.7.2)$$

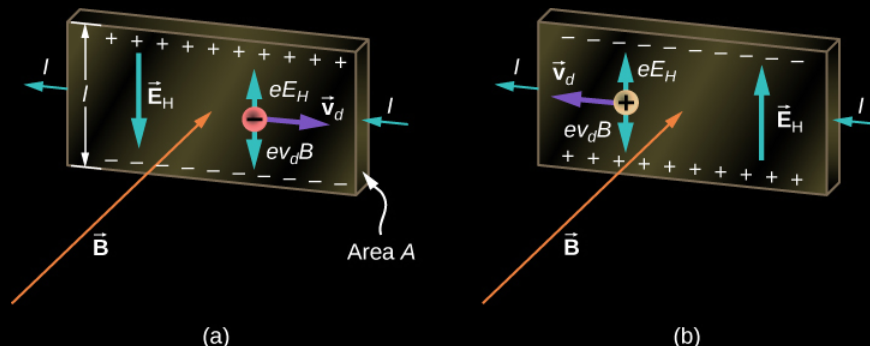


Figure 11.7.1: In the Hall effect, a potential difference between the top and bottom edges of the metal strip is produced when moving charge carriers are deflected by the magnetic field. (a) Hall effect for negative charge carriers; (b) Hall effect for positive charge carriers.

A scenario where the electric and magnetic fields are perpendicular to one another is called a crossed-field situation. If these fields produce equal and opposite forces on a charged particle with the velocity that equates the forces, these particles are able to pass through an apparatus, called a **velocity selector**, undeflected. This velocity is represented in Equation 11.7.3. Any other velocity of a charged particle sent into the same fields would be deflected by the magnetic force or electric force.

Going back to the Hall effect, if the current in the strip is I , then from **Current and Resistance**, we know that

$$I = nev_d A \quad (11.7.3)$$

where n is the number of charge carriers per volume and A is the cross-sectional area of the strip. Combining the equations for v_d and I results in

$$I = ne \left(\frac{E}{B} \right) A. \quad (11.7.4)$$

The field E is related to the potential difference V between the edges of the strip by

$$E = \frac{V}{l}. \quad (11.7.5)$$

The quantity V is called the **Hall potential** and can be measured with a voltmeter. Finally, combining the equations for I and E gives us

$$V = \frac{IBl}{neA} \quad (11.7.6)$$

where the upper edge of the strip in Figure 11.7.1 is positive with respect to the lower edge.

We can also combine Equation 11.7.1 and Equation 11.7.5 to get an expression for the Hall voltage in terms of the magnetic field:

$$V = Blv_d. \quad (11.7.7)$$

What if the charge carriers are positive, as in Figure 11.7.1? For the same current I , the magnitude of V is still given by Equation 11.7.6. However, the upper edge is now negative with respect to the lower edge. Therefore, by simply measuring the sign of V , we can determine the sign of the majority charge carriers in a metal.

Hall potential measurements show that electrons are the dominant charge carriers in most metals. However, Hall potentials indicate that for a few metals, such as tungsten, beryllium, and many semiconductors, the majority of charge carriers are positive. It turns out that conduction by positive charge is caused by the migration of missing electron sites (called holes) on ions. Conduction by holes is studied later in **Condensed Matter Physics**.

The Hall effect can be used to measure magnetic fields. If a material with a known density of charge carriers n is placed in a magnetic field and V is measured, then the field can be determined from Equation 11.7.7. In research laboratories where the fields of electromagnets used for precise measurements have to be extremely steady, a “Hall probe” is commonly used as part of an electronic circuit that regulates the field.

✓ Example 11.7.1: Velocity Selector

An electron beam enters a crossed-field velocity selector with magnetic and electric fields of 2.0 mT and $6.0 \times 10^3 \text{ N/C}$, respectively. (a) What must the velocity of the electron beam be to traverse the crossed fields undeflected? If the electric field is turned off, (b) what is the acceleration of the electron beam and (c) what is the radius of the circular motion that results?

Strategy

The electron beam is not deflected by either of the magnetic or electric fields if these forces are balanced. Based on these balanced forces, we calculate the velocity of the beam. Without the electric field, only the magnetic force is used in Newton’s second law to find the acceleration. Lastly, the radius of the path is based on the resulting circular motion from the magnetic force.

Solution

1. The velocity of the unperturbed beam of electrons with crossed fields is calculated by Equation 11.7.2:

$$v_d = \frac{E}{B} = \frac{6 \times 10^3 \text{ N/C}}{2 \times 10^{-3} \text{ T}} = 3 \times 10^6 \text{ m/s}. \quad (11.7.8)$$

2. The acceleration is calculated from the net force from the magnetic field, equal to mass times acceleration. The magnitude of the acceleration is:

$$ma = qvB \quad (11.7.9)$$

$$a = \frac{qvB}{m} = \frac{(1.6 \times 10^{-19} \text{ C})(3 \times 10^6 \text{ m/s})(2 \times 10^{-3} \text{ T})}{0.1 \times 10^{-31} \text{ kg}} = 1.1 \times 10^{15} \text{ m/s}^2. \quad (11.7.10)$$

3. The radius of the path comes from a balance of the circular and magnetic forces, or Equation 11.7.2

$$r = \frac{mv}{qB} = \frac{(9.1 \times 10^{-31} \text{ kg})(3 \times 10^6 \text{ m/s})}{(1.6 \times 10^{-19} \text{ C})(2 \times 10^{-3} \text{ T})} = 8.5 \times 10^{-3} \text{ m}. \quad (11.7.11)$$

Significance

If electrons in the beam had velocities above or below the answer in part (a), those electrons would have a stronger net force exerted by either the magnetic or electric field. Therefore, only those electrons at this specific velocity would make it through.

✓ The Hall Potential in a Silver Ribbon

Figure 11.7.2 shows a silver ribbon whose cross section is 1.0 cm by 0.20 cm. The ribbon carries a current of 100 A from left to right, and it lies in a uniform magnetic field of magnitude 1.5 T. Using a density value of $n = 5.9 \times 10^{28}$ electrons per cubic meter for silver, find the Hall potential between the edges of the ribbon.

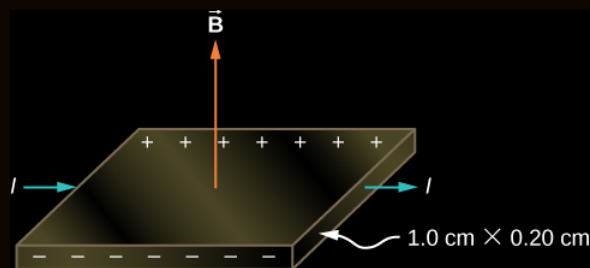


Figure 11.7.2: Finding the Hall potential in a silver ribbon in a magnetic field is shown.

Strategy

Since the majority of charge carriers are electrons, the polarity of the Hall voltage is that indicated in the figure. The value of the Hall voltage is calculated using Equation 11.7.6

Solution

When calculating the Hall voltage, we need to know the current through the material, the magnetic field, the length, the number of charge carriers, and the area. Since all of these are given, the Hall voltage is calculated as:

$$\begin{aligned} v &= \frac{IBl}{neA} \\ &= \frac{(100 \text{ A})(1.5 \text{ T})(1.0 \times 10^{-2} \text{ m})}{(5.9 \times 10^{28} / \text{m}^3)(1.6 \times 10^{-19} \text{ C})(2.0 \times 10^{-5} \text{ m}^2)} \\ &= 7.9 \times 10^{-6} \text{ V}. \end{aligned}$$

Significance

As in this example, the Hall potential is generally very small, and careful experimentation with sensitive equipment is required for its measurement.

? Exercise 11.7.1

A Hall probe consists of a copper strip, $n = 8.5 \times 10^{28}$ electrons per cubic meter, which is 2.0 cm wide and 0.10 cm thick. What is the magnetic field when $I = 50$ A and the Hall potential is

- $4.0 \mu\text{V}$ and
- $6.0 \mu\text{V}$?

Answer a

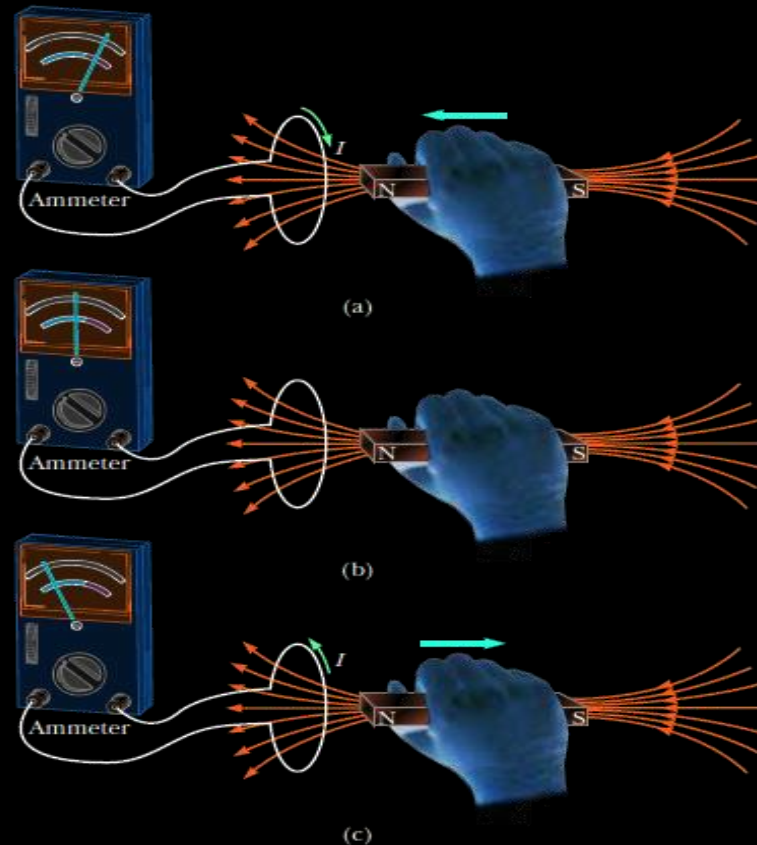
1.1 T

Answer b

1.6 T

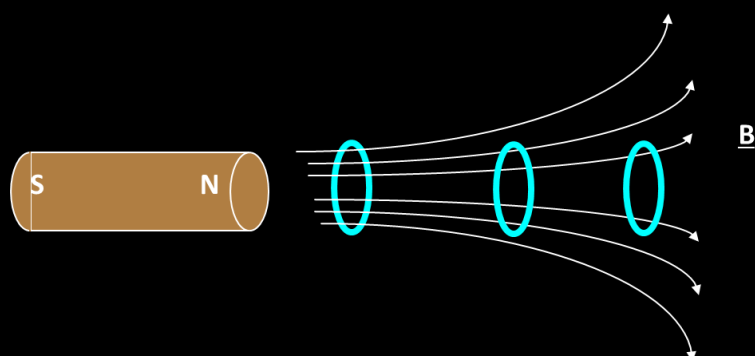
Electromagnetic Induction:

The process of developing electromotive force and current in a closed coil with the help of a moving magnet or current carrying coil is called electromotive force.



Faraday's Law:

Frist Law: Whenever there is a change in the number of magnetic field lines or magnetic flux passing through a closed coil then an electromotive force or electric current is induced in the coil. The induced electromotive force or electric current exists as long as the magnetic field lines or magnetic flux keep changing.



Second Law: The magnitude of the induced electromotive force in a closed coil is directly proportional to the negative of the rate of change of magnetic flux passing through the coil.

$$\epsilon \propto - \frac{d\phi}{dt}$$

$$\Leftrightarrow \epsilon = -K \frac{d\phi}{dt}$$

The value of proportionality constant is 1 in SI unit. So can write,

$$\epsilon = - \frac{d\phi}{dt}$$

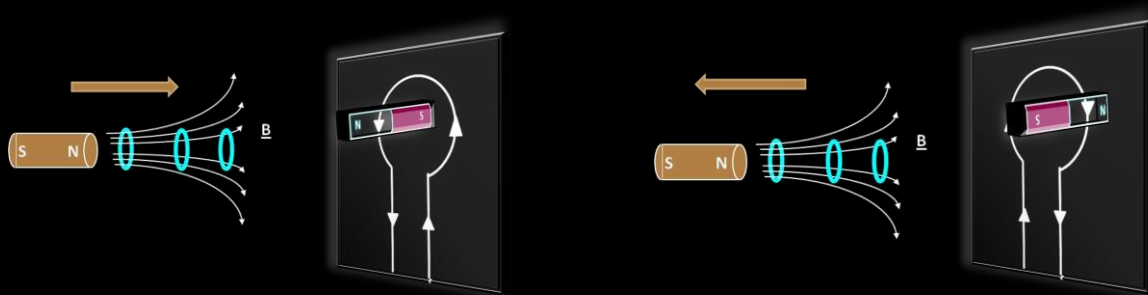
For N number of turns in the coil,

$$\epsilon = -N \frac{d\phi}{dt}$$

It is also known as the Newman's Law.

Lenz's Law:

"The direction of induced electromotive force or electric current is such that it always tends to oppose the very cause for which it is produced".



a) Anti-Clockwise current direction

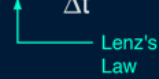
b) Clockwise current direction

When a N-pole of a magnet toward the coil, the electric current is induced and directed to anti-clockwise. So that it acts "in opposition" and tends to oppose the direction of the magnet (Fig. a). In the same way, when N-pole away from the coil, it also acts "in opposition" and tends to oppose the magnet movement and the direction of the induced current will be clockwise (Fig. b).

So that, the induced emf in a coil is equal to the negative of the rate of change of magnetic flux times the number of turns in the coil.

Faraday's Law

$$\text{Emf} = -N \frac{\Delta\Phi}{\Delta t}$$


Lenz's Law

where N = number of turns
 $\Phi = BA$ = magnetic flux
 B = external magnetic field
 A = area of coil

The minus sign indicated the Lenz's law.

Explanation of “Lenz law is in accordance with the law of conservation of energy”

“To move the magnet towards the coil, mechanical work has to be done to overcome the force of repulsion between the north poles of the bar magnet and the coil. This mechanical work done is converted into electrical energy (indicated by the deflection in the galvanometer). Similarly, when the magnet moves away from the coil, the upper face of the coil acquires south polarity. In this case, the induced e.m.f. will oppose the outward motion (cause) of the magnet. Again mechanical work has to be done to overcome the force of attraction between north pole of the magnet and south pole of the coil. This work done is converted as electrical energy. If the magnet is not moved, no mechanical work is done, then no e.m.f. (thus no current) is induced in the coil i.e., no electrical energy is produced. Thus, Lenz law is in accordance with the law of conservation of energy”.

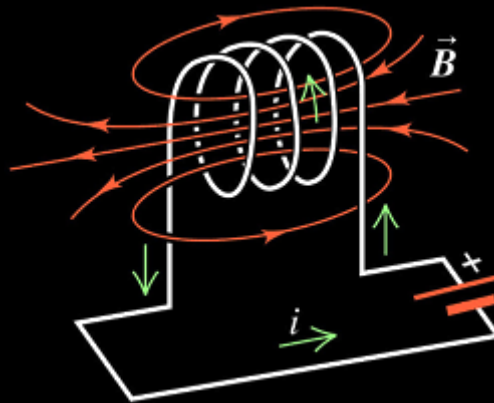
Types of Inductance:

Two types of inductance are there:

- Self-Induction
- Mutual Induction

What is Self-Induction?

When there is a change in the current or magnetic flux of the coil, an opposed induced electromotive force is produced. This phenomenon is termed as Self Induction. When the current starts flowing through the coil at any instant, it is found that, that the magnetic flux becomes directly proportional to the current passing through the circuit.



The relation is given as:

$$\begin{aligned} \phi &\propto i \\ \Leftrightarrow \quad \phi &= Li \end{aligned}$$

Where L is termed as self-inductance of the coil or the coefficient of self-inductance. The self-inductance depends on the cross-sectional area, the permeability of the material, or the number of turns in the coil.

The rate of change of magnetic flux in the coil is given as,

$$\begin{aligned} \epsilon &= -\frac{d\phi}{dt} = -\frac{d(Li)}{dt} \\ \Leftrightarrow \quad \epsilon &= -L \frac{di}{dt} \end{aligned}$$

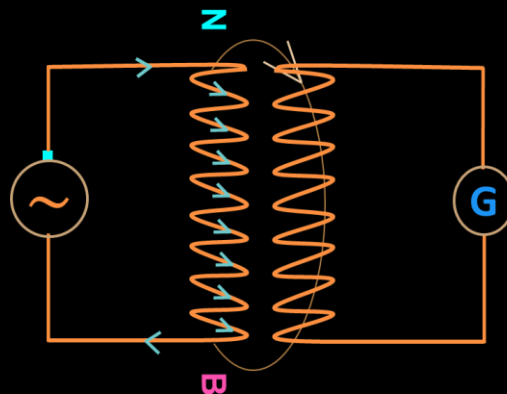
The formula of self-inductance is –

$$L = -\frac{\epsilon}{\left(\frac{di}{dt}\right)} \quad \text{or} \quad L = N \frac{\phi}{i}$$

The unit of the self-inductance is Henry.

What is Mutual Induction?

We take two coils, and they are placed close to each other. The two coils are P- coil (Primary coil) and S- coil (Secondary coil). To the P-coil, a battery, and a key is connected wherein the S-coil a galvanometer is connected across it. When there is a change in the current or magnetic flux linked with two coils an opposing electromotive force is produced across each coil, and this phenomenon is termed as Mutual Induction.



The relation is given as:

$$\begin{aligned}\phi &\propto i \\ \Leftrightarrow \quad \phi &= Mi\end{aligned}$$

Where M is termed as the mutual inductance of the two coils or the coefficient of the mutual inductance of the two coils.

The rate of change of magnetic flux in the coil is given as,

$$\begin{aligned}\epsilon &= -\frac{d\phi}{dt} = -\frac{d(Mi)}{dt} \\ \Leftrightarrow \quad \epsilon &= -M \frac{di}{dt}\end{aligned}$$

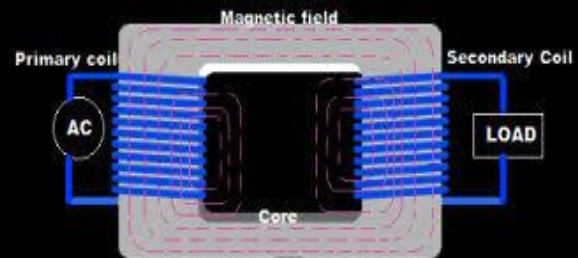
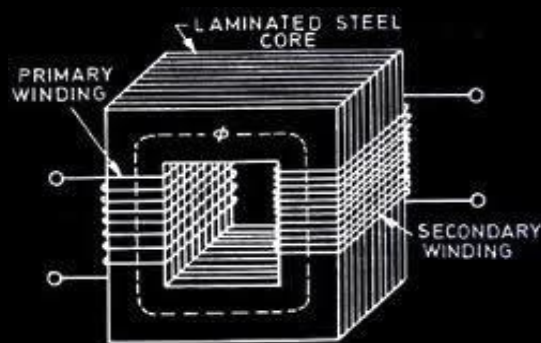
The formula of mutual inductance is –

$$M = -\frac{\epsilon}{\left(\frac{di}{dt}\right)} \quad \text{or} \quad M = N \frac{\phi}{i}$$

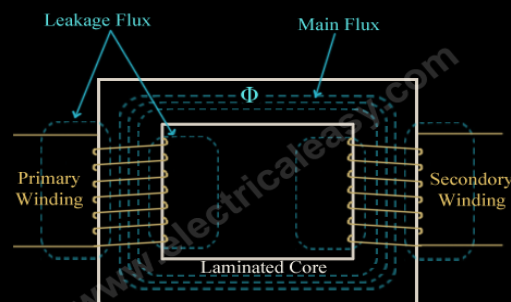
The unit of the mutual inductance is Henry.

Transformer:

Electrical transformer is a static electrical machine which transforms electrical power from one circuit to another circuit, without changing the frequency. Transformer can increase or decrease the voltage with corresponding decrease or increase in current.



- **Working principle of transformer:** The basic principle behind working of a transformer is the phenomenon of mutual induction between two windings linked by common magnetic flux. The figure at right shows the simplest form of a transformer. Basically, a transformer consists of two inductive coils; primary winding and secondary winding. The coils are electrically separated but magnetically linked to each other.



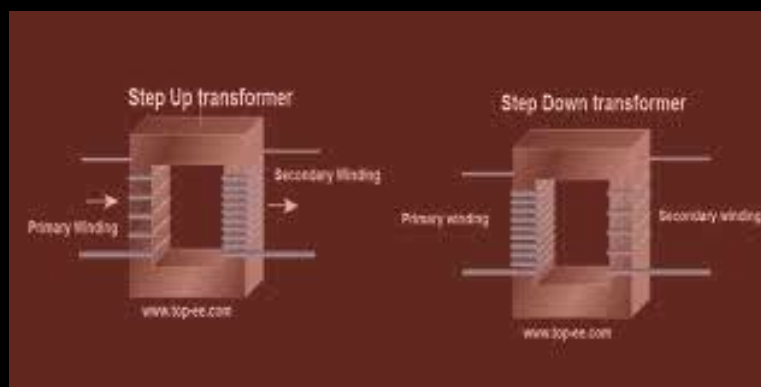
When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding. The core provides magnetic path for the flux, to get linked with the secondary winding. Most of the flux gets linked with the secondary winding which is called as 'useful flux' or main 'flux', and the flux which does not get linked with secondary winding is called as 'leakage flux'. As the flux produced is alternating (the direction of it is continuously changing), EMF gets induced in the secondary winding according to Faraday's law of electromagnetic induction. This emf is called 'mutually induced

emf, and the frequency of mutually induced emf is same as that of supplied emf. If the secondary winding is closed circuit, then mutually induced current flows through it, and hence the electrical energy is transferred from one circuit (primary) to another circuit (secondary).

Formula:

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \quad \Leftrightarrow \quad \frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{I_s}{I_p} \quad \Leftrightarrow \quad \frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Types of Transformer:



- **Step-up Transformer:**

A transformer in which the output (secondary) voltage is greater than its input (primary) voltage is called a step-up transformer.

In this type of transformer, the number of turns on the secondary coil is greater than that of the primary.

- **Step-down Transformer**

A transformer in which the output (secondary) voltage is less than its input (primary) voltage is called a step-down transformer. In this type of transformer, the number of turns on the primary coil is greater than the turn on the secondary coil of the transformer.

Home works:

- Define 1 Henry
- Define self-inductance and mutual inductance.
- Write down the basic differences between the step up and step down transformer.

Direct and Alternative current:

Most of the examples dealt with so far, and particularly those utilizing batteries, have constant voltage sources. Once the current is established, it is thus also a constant. *Direct current (DC) is the flow of electric charge in only one direction. It is the steady state of a constant-voltage circuit.*

Most well-known applications, however, use a time-varying voltage source. *Alternating current (AC) is the flow of electric charge that periodically reverses direction.* If the source varies periodically, particularly sinusoidally, the circuit is known as an alternating current circuit. Examples include the commercial and residential power that serves so many of our needs. Figure 1 shows graphs of voltage and current versus time for typical DC and AC power. The AC voltages and frequencies commonly used in homes and businesses vary around the world.

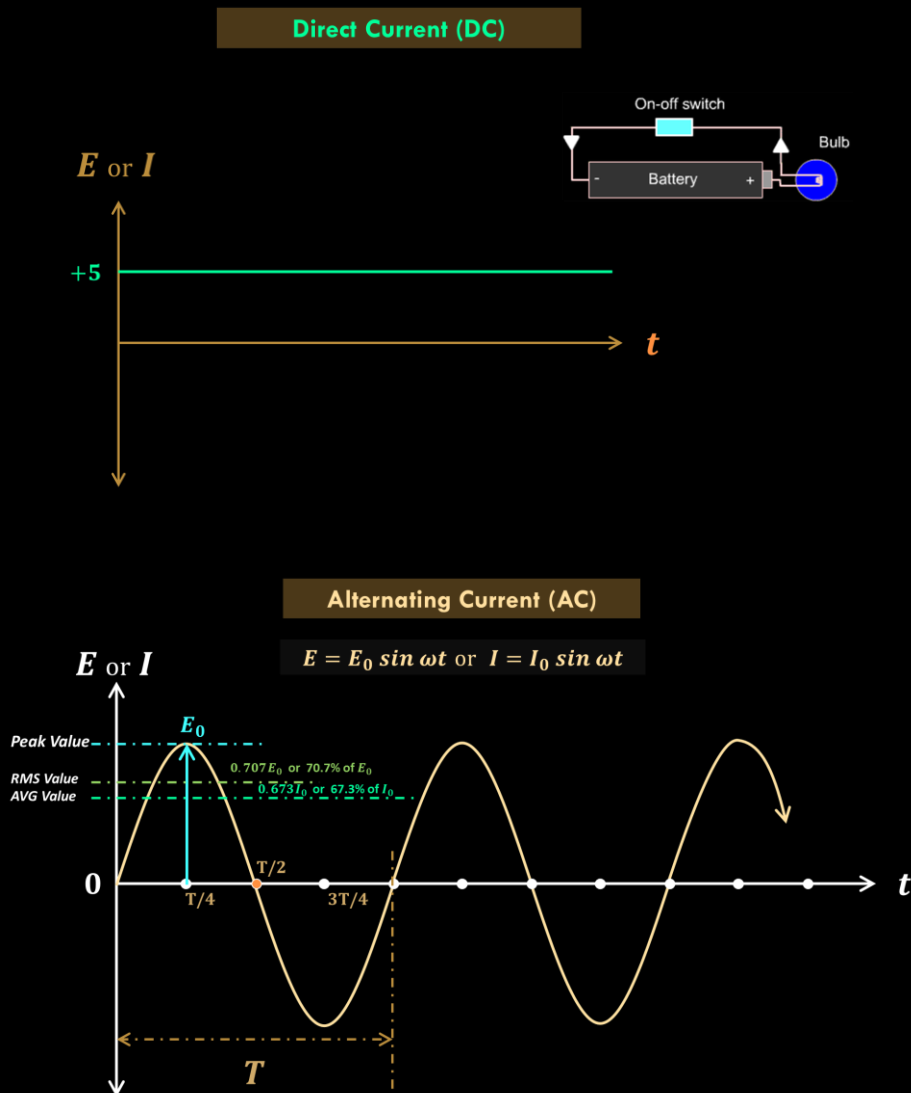


Figure 2 shows a schematic of a simple circuit with an AC voltage source. The voltage between the terminals fluctuates as shown, with the *AC voltage* given by

$$V = V_0 \sin \omega t$$

where V is the voltage at time t and V_0 is the peak voltage. For this simple resistance circuit, $I=V/R$, and so the *AC current* is

$$I = I_0 \sin \omega t$$

Common Formula of AC:

- Peak value, V_0 or E_0 and I_0
- Angular velocity or angular frequency, $\omega = 2\pi f$ or $\omega = \frac{2\pi}{T}$
- Frequency, $f = \frac{1}{T}$ or $f = \frac{\omega}{2\pi}$
- Time period, $T = \frac{1}{f}$ or $T = \frac{2\pi}{\omega}$
- Average value for full cycle, $I_{avg} = 0$ or $E_{avg} = 0$
- Average value for half cycle, $I_{avg} = 0.637I_0$ or $E_{avg} = 0.637E_0$
- Root Mean Square (RMS) value for full cycle, $I_{rms} = 0.707I_0$ or $E_{rms} = 0.707E_0$
- Form Factor = $\frac{rms\ value}{avg\ value} = 1.11$